

The Fundamental Theorem of Calculus

ANSWER KEY



FTC Part 2: evaluating definite integrals

- 1 Evaluate $\int_0^3 (x^2 - 2x) dx$ using the Fundamental Theorem of Calculus.
 $F(x) = \frac{x^3}{3} - x^2$. $F(3) - F(0) = (9 - 9) - (0) = 0$.
- 2 Evaluate $\int_1^4 \sqrt{x} dx$.
 $F(x) = \frac{2}{3}x^{3/2}$. $F(4) - F(1) = \frac{2}{3}(8) - \frac{2}{3}(1) = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$.
- 3 Evaluate $\int_0^{\pi/2} \cos x dx$.
 $F(x) = \sin x$. $F(\pi/2) - F(0) = 1 - 0 = 1$.

FTC Part 1: accumulation functions

- 4 Let $g(x) = \int_0^x t^2 dt$. Find $g'(x)$ using FTC Part 1.
By FTC Part 1, $g'(x) = x^2$ (the integrand evaluated at x).
- 5 Let $g(x) = \int_1^x \sin(t^2) dt$. Find $g'(2)$.
 $g'(x) = \sin(x^2)$, so $g'(2) = \sin(4) \approx -0.757$.
- 6 Let $g(x) = \int_0^{x^2} e^{-t} dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.
 $g'(x) = e^{-x^2} \cdot \frac{d}{dx}(x^2) = 2xe^{-x^2}$.

Interpreting accumulation functions

- 7 A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau) d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.
 $s(4) = \int_0^4 (3 - 2\tau) d\tau = [3\tau - \tau^2]_0^4 = (12 - 16) - 0 = -4$. This means the particle's net displacement from $t = 0$ to $t = 4$ is -4 m (it ends up 4 m to the left/behind its starting point).
- 8 Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t) dt$, explain what feature of g 's graph occurs where f crosses the x -axis, and where f has a local max.
Where f crosses the x -axis (changes sign), $g' = f$ changes sign, so g has a local extremum there. Where f has a local max, $g' = f$ has a local max, meaning g has an inflection point there (since $g'' = f'$ changes sign from $+$ to $-$).

Average value of a function

- 9 Find the average value of $f(x) = x^2$ on $[0, 3]$.

$$\text{Average} = \frac{1}{3-0} \int_0^3 x^2 dx = \frac{1}{3} \left[\frac{x^3}{3} \right]_0^3 = \frac{1}{3}(9) = 3.$$

- 10 The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$. Find the average temperature over this period.

Average $= \frac{1}{24} \int_0^{24} \left[70 + 10\sin\left(\frac{\pi t}{12}\right) \right] dt$. The sine term integrates to 0 over a full period (24 hours = 2 full cycles), so the average $= 70^\circ$.

Definite integrals with substitution-friendly integrands

- 11 Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$ using $u = x^2 + 1$.

$$du = 2x dx. \text{ When } x = 0, u = 1; \text{ when } x = 1, u = 2. \int_1^2 u^3 du = \left[\frac{u^4}{4} \right]_1^2 = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}.$$

The Mean Value Theorem for integrals

- 12 Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

The average value of $f(x) = x^2$ on $[0, 3]$ is 3 (found earlier). Set $f(c) = c^2 = 3$: $c = \sqrt{3} \approx 1.73$, which lies in $(0, 3)$.

Visualizing FTC Part 1 as an accumulation function

- 13 The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2 dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x .

Since $f(t) = t^2 \geq 0$ for all t , every strip added as x increases contributes non-negative area, so $g(x) = \int_0^x t^2 dt$ is always increasing (or constant only at $x = 0$). By FTC Part 1, $g'(x)$ equals the height of the integrand curve at x , i.e. $g'(x) = f(x) = x^2$, matching the graph's height directly.

A capstone FTC application

- 14 Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \leq t \leq 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

Volume $= \int_0^6 (8 - t) dt = \left[8t - \frac{t^2}{2} \right]_0^6 = 48 - 18 = 30$ liters. Since $r(t) = 8 - t = 0$ at $t = 8$, the inflow rate drops to zero exactly when the tank presumably reaches capacity, consistent with the water level leveling off.

FTC Part 1 with more complex bounds

- 15 Let $g(x) = \int_3^{2x} \ln(t+1) dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.
 $g'(x) = \ln(2x+1) \cdot \frac{d}{dx}(2x) = 2\ln(2x+1)$.
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A second average-value application

- 16 A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \leq t \leq 8$. Find the average production rate over the shift.

Average

$$= \frac{1}{8} \int_0^8 (50 + 10t - t^2) dt = \frac{1}{8} \left[50t + 5t^2 - \frac{t^3}{3} \right]_0^8 = \frac{1}{8} \left(400 + 320 - \frac{512}{3} \right) = \frac{1}{8} \left(\frac{2160 - 512}{3} \right) = \frac{1648}{24} \approx 68.67$$

units/hour.